Code No: 121AH **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year Examinations, March/April - 2023
ENGINEERING DRAWING
(Common to CSE, MIE)

Time: 3 hours **Max Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Construct a vernier scale to read the distances corrected to decameter on a map in which the actual distances are reduced in the ratio of 1:50,000. The scale should be long enough to measure 8 km. Mark a length of 4.82 km on scale by constructing backward vernier scales.

b) A circus man rides a motor-cycle inside a globe of 4 m diameter. The motorcycle wheel diameter is 0.8 m. Draw the locus of a point spot on the circumference of the motor-cycle wheel for its one complete turn. Adopt a suitable scale. [7+8]

OR

2.a) The ordinate of a point P on the curve is 40 mm and is at a distance of 20 mm from the vertex. Draw the parabola.

b) Draw a diagonal scale of RF = 1/20,000 to show kilometres and decimals of kilometre. Mark on the scale the distances 1.64 km and 2.37 km. [7+8]

3. The projections of a line AB are perpendicular to xy. The end A is in the HP and 50 mm in front of the VP and the end B is in the VP and 40 mm above the HP. Draw its projections, determine its true length and the inclinations with the reference planes. [15]

OR

4. The top view '*abc*' of a triangle *ABC* is an equilateral triangle of side 50 mm, *ab* being inclined at 45^0 to *xy*. The point *A* is in the V.P. and 35 mm above the ground, while the points *B* and *C* are on the ground. Draw the projections of the triangle and determine its true shape. [15]

5. A cylinder of base diameter 50 mm and axis 70 mm is lying on a generator on the HP with its axis parallel to the VP. It is cut by an A.P. inclined at 30^0 to the HP and passing through a point on the axis 30 mm from one of its ends. Draw its sectional top view and obtain true shape of the section. [15]

OR

6. A pentagonal pyramid, side of base 35 mm and axis 70 mm long, is lying on one of its corners on the HP such that the triangular surface opposite to the corner is parallel to HP and perpendicular to VP, and the base edge containing that triangular surface is parallel to both the HP and VP. Draw the projections of the solid when the apex of the pyramid is nearer to the observer. [15]

7. A cone of 100 mm diameter and axis 120 mm long, resting on its base in the HP is penetrated by a horizontal cylinder 50 mm diameter and 120 mm long. The axis of the cylinder is 30 mm above the base of the cone and parallel to the VP. If the axis of the cylinder is 5 mm in front of the axis of the cone, draw the projections of the solids showing the curves of intersection. [15]

OR

8. A cuboid of sides 40 mm × 60 mm × 40 mm is resting on its longer side in the H.P. and one its longer surfaces parallel to the V.P. It is cut by a section plane perpendicular to V.P. and incline to the H.P. such that it passes diagonally from one corner of the top to the opposite corner of the bottom. Develop the lateral surface of the bottom part of cuboid. [15]

9. Draw top, front and side views of the isometric projection given in Figure.1. All dimensions are in mm. [15]

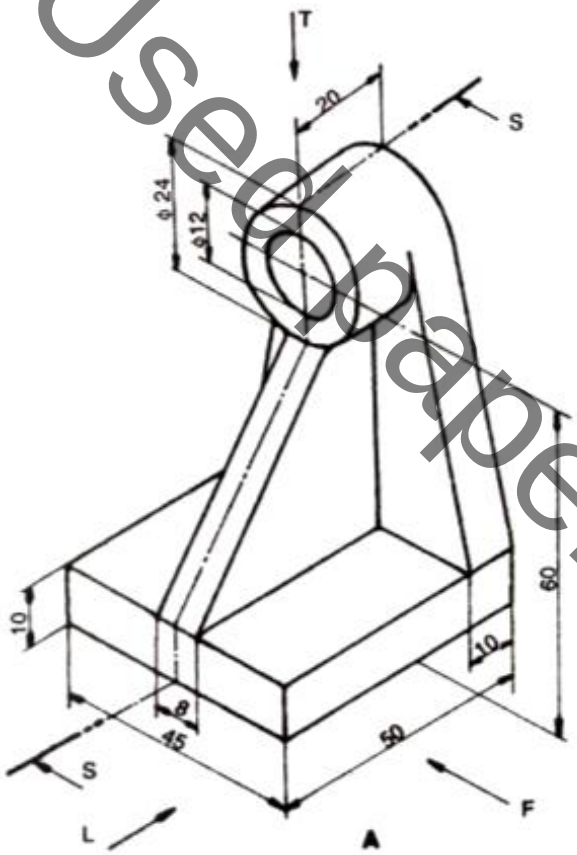


Figure 1

OR

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10. Draw the isometric view of the object whose orthographic projections are given in Figure.2. All dimensions are in mm. [15]

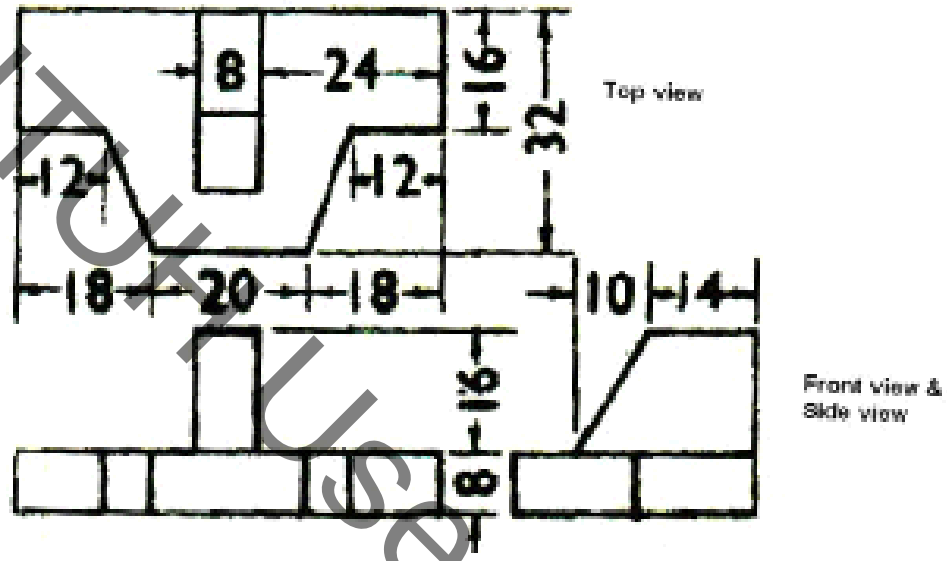


Figure 2

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